Electrocortical Responses in Anticipation of Avoidable and Inevitable Threats: A Multisite Study

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Abstract

When faced with danger, human beings respond with a repertoire of defensive behaviors, including freezing and active avoidance. Previous research has revealed a pattern of physiological responses, characterized by heart rate bradycardia, reduced visual exploration, and heightened sympathetic arousal in reaction to avoidable threats, suggesting a state of attentive immobility in humans. However, the electrocortical underpinnings of these behaviors remain largely unexplored. To investigate the visuocortical components of attentive immobility, we recorded parieto-occipital alpha activity, along with eye movements and autonomic responses, while participants awaited either an avoidable, inevitable, or no threat. To test the robustness and generalizability of our findings, we collected data from a total of 101 participants (76 females, 25 males) at two laboratories. Across sites, we observed an enhanced suppression of parieto-occipital alpha activity during avoidable threats, in contrast to inevitable or no threat trials, particularly toward the end of the trial that prompted avoidance responses. This response pattern coincided with heart rate bradycardia, centralization of gaze, and increased sympathetic arousal. Furthermore, our findings expand on previous research by revealing that the amount of alpha suppression, along with centralization of gaze, and heart rate changes predict the speed of motor responses. Collectively, these findings indicate that when individuals encounter avoidable threats, they enter a state of attentive immobility, which enhances perceptual processing and facilitates action preparation. This state appears to reflect freezing-like behavior in humans.